Lectures 21-22a

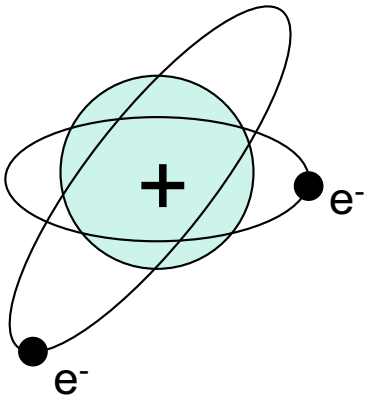
Plane Waves in Materials

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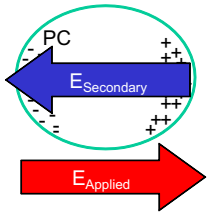
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3 types of materials

Conductors



Free electrons



$\vec{E} = 0$ inside

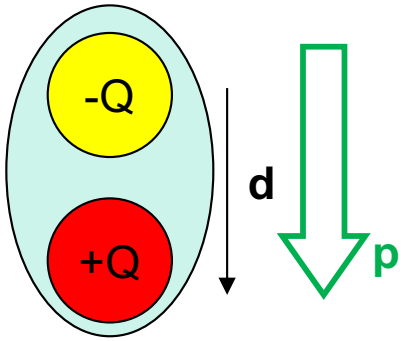
$\rho = 0$ inside

$\rho = \rho_s$ only surface charge

V is same throughout

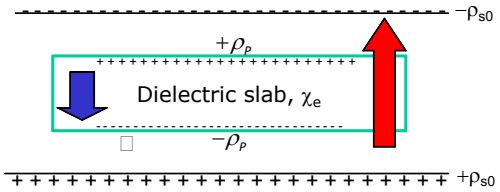
$\vec{E}_{outside}$ is $\perp$ to surface

Dielectrics



Polarized atoms/molecules

Bound electrons

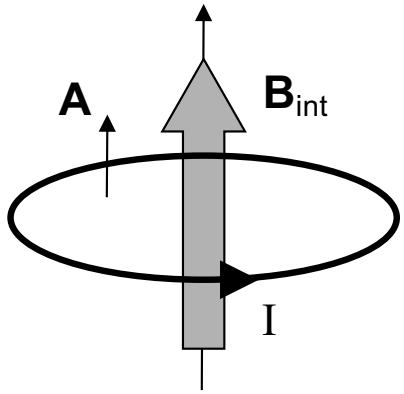


$\vec{E} \neq 0$ inside but it is reduced

$$\vec{E}_{tot} = \vec{E}_0 + \vec{E}_{\text{int}}$$

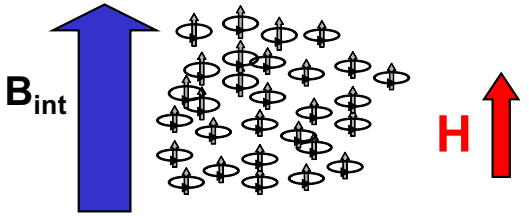
$$\vec{D} = \epsilon\vec{E}_{\text{tot}} = \vec{P} + \epsilon_0\vec{E}_{tot}$$

Magnetic



Magnetic moments

Bound electrons



$\vec{H} \neq 0$ inside but it can be reduced or increased

$$\vec{B}_{tot} = \vec{B}_0 + \vec{B}_{\text{int}}$$

$$\vec{B}_{tot} = \mu\vec{H} = \mu_0(\vec{H} + \vec{M})$$

Constitution Relations

conduction $\vec{J} = \sigma \vec{E}$

polarization $\vec{\boldsymbol{P}} = \epsilon_0 \chi_e \vec{\boldsymbol{E}}$ $\vec{D} = \varepsilon \vec{E}$ $\varepsilon = \varepsilon_0 \varepsilon_r$

magnetization $\vec{\boldsymbol{M}} = \chi_m \vec{\boldsymbol{H}}$ $\vec{B} = \mu \vec{H}$ $\mu = \mu_0 \mu_r$

Inside a material, Maxwell's Equations become:

Free Space	Inside Material
$\nabla \times \vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t}$	$\nabla \times \vec{H} = \sigma \vec{E} + \varepsilon \frac{\partial \vec{E}}{\partial t}$
$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$	$\nabla \times \vec{E} = -\mu \frac{\partial \vec{H}}{\partial t}$
$\nabla \bullet \vec{D} = \rho$	$\nabla \bullet (\varepsilon \vec{E}) = \rho$
$\nabla \bullet \vec{B} = 0$	$\nabla \bullet \vec{H} = 0$

A single material can have conductive, dielectric, and magnetic properties AT THE SAME TIME

$$\sigma \neq 0$$

$$\varepsilon_r \neq 1$$

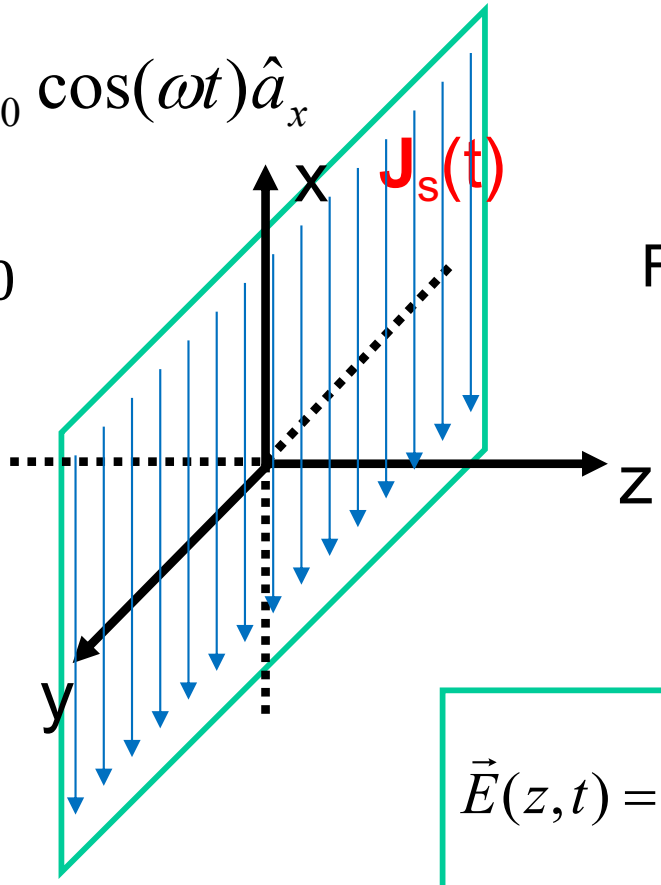
$$\mu_r \neq 1$$

How does an EM wave propagate through this?

Recall: Field Source

Infinite plane sheet of current at z=0

$$\vec{J}_s = -J_{S0}\cos(\omega t)\hat{a}_x$$



Free space $z < 0$

Free space $z > 0$

J_s = surface current [A/m] at $z = 0$

Solution

$$\vec{E}(z,t) = \frac{\eta_0 J_{S0}}{2}\cos(\omega t \mp \beta z)\hat{a}_x \quad z \gtrless 0$$

$$\vec{H}(z,t) = \pm\frac{J_{S0}}{2}\cos(\omega t \mp \beta z)\hat{a}_y \quad \beta = \frac{\omega}{v_p}$$

Recall: Key characteristics in FREE SPACE

- No attenuation
- E_x and H_y are IN PHASE for linear polarization
- Impedance: $\left|\vec{E}\right| = \eta_0 \left|\vec{H}\right|$
- Travel at Speed of Light: $v_p = c = \omega/\beta$
- Perpendicular: **E** ⊥ **H** ⊥ **S**

$$\vec{S} = \vec{E} \times \vec{H}$$

Solution

$$\vec{E}(z,t) = \frac{\eta_0 J_{S0}}{2} \cos(\omega t \mp \beta z)\hat{a}_x \quad z \gtrless 0$$

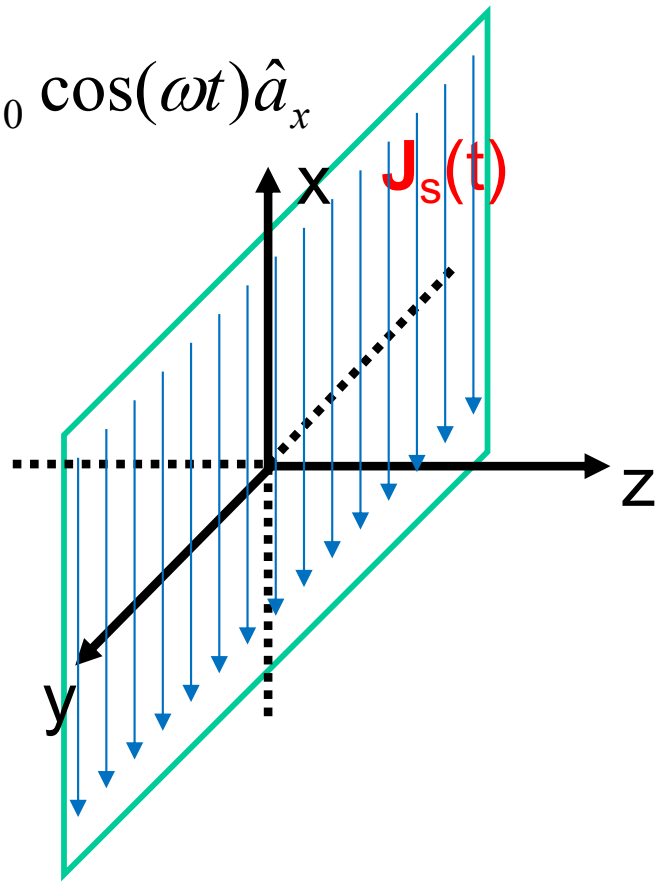
$$\vec{H}(z,t) = \pm \frac{J_{S0}}{2} \cos(\omega t \mp \beta z)\hat{a}_y \quad \beta = \frac{\omega}{v_p}$$

7

Field Source in a MATERIAL

Infinite plane sheet of current at z=0

$$\vec{J}_s = -J_{S0}\cos(\omega t)\hat{a}_x$$



Material z<0

σ, μ, ε

Material z>0

σ, μ, ε

J_s = surface current [A/m] at $z = 0$

Similar Procedure for Maxwell's Equations

- We will SIMULTANEOUSLY solve Maxwell's equations to find $\vec{E}$ and $\vec{H}$ caused by $\vec{J}$

  - Note: $\vec{J} = \sigma\vec{E} \neq 0$ inside the material

$$\vec{\nabla} \times \vec{E} = -\mu \frac{\partial \vec{H}}{\partial t}$$

$$\vec{\nabla} \times \vec{H} = \sigma\vec{E} + \varepsilon \frac{\partial \vec{E}}{\partial t}$$

$$\vec{J}_S = -J_x(t)\hat{a}_x$$

$$\vec{E} = E_x(z,t)\hat{a}_x$$

$$\vec{H} = H_y(z,t)\hat{a}_y$$

Phasor Review

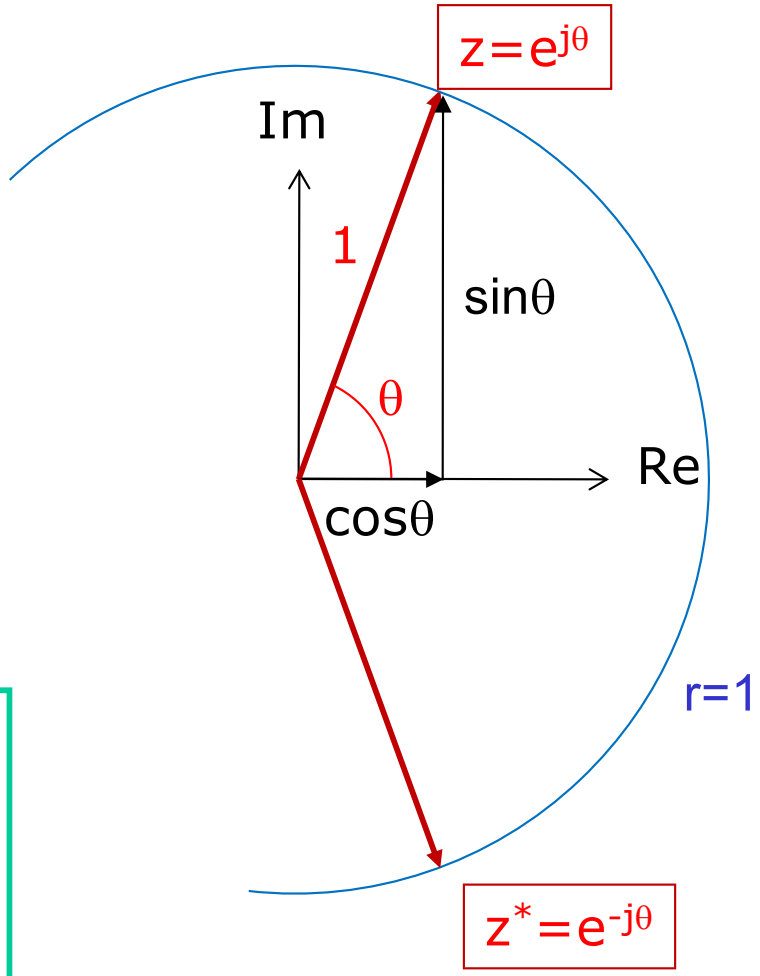
$$e^{j\theta} = \cos\theta + j\sin\theta$$

$$\cos(\theta) = \mathrm{Re}[e^{j\theta}]$$

$$\sin(\theta) = \mathrm{Re}[-je^{j\theta}] = \mathrm{Re}[e^{j(\theta-\pi/2)}]$$

$$\mathrm{Re}[z] = (z + z^*)/2$$

$$\begin{aligned} E_x(z,t) &= E_x(z)\cos(\omega t \mp \beta z + \theta) \\ &= \mathrm{Re}[E_x(z)e^{\mp j\beta z}e^{j\theta}e^{j\omega t}] \\ &= \mathrm{Re}[\tilde{E}_x(z)e^{j\omega t}] \end{aligned}$$



Solve PDEs with Phasors

Technique simplifies the algebra

$$E_x(z,t) = \mathrm{Re}[\tilde{E}_x(z)e^{j\omega t}] \qquad H_y(z,t) = \mathrm{Re}[\tilde{H}_y(z)e^{j\omega t}]$$

$$\frac{\partial E_x}{\partial t} = \mathrm{Re}[j\omega\tilde{E}_x(z)e^{j\omega t}] \qquad \boxed{\frac{\partial}{\partial t} \equiv j\omega}$$

$$\left.\begin{aligned} \frac{\partial E_x}{\partial z} &= -\mu\frac{\partial H_y}{\partial t} \\ \frac{\partial H_y}{\partial z} &= -\sigma E_x - \varepsilon\frac{\partial E_x}{\partial t} \end{aligned}\right\} \quad \begin{aligned} \frac{\partial \tilde{E}_x}{\partial z} &= -\mu(j\omega)\tilde{H}_y \\ \frac{\partial \tilde{H}_y}{\partial z} &= -\sigma\tilde{E}_x - \varepsilon(j\omega)\tilde{E}_x \end{aligned}$$

Wave Equation

$$\frac{\partial \tilde{E}_x}{\partial z} = -\mu(j\omega)\tilde{H}_y$$

$$\frac{\partial \tilde{H}_y}{\partial z} = -\sigma\tilde{E}_x - \varepsilon(j\omega)\tilde{E}_x$$

Differentiate again in z

$$\frac{\partial^2 \tilde{E}_x}{\partial z^2} = -\mu(j\omega)\frac{\partial \tilde{H}_y}{\partial z}$$

$$= -\mu(j\omega)[-\sigma\tilde{E}_x - \varepsilon(j\omega)\tilde{E}_x]$$

$$= j\omega\mu(\sigma + j\omega\varepsilon)\tilde{E}_x$$

$$\boxed{\frac{\partial^2 \tilde{E}_x}{\partial z^2} = \bar{\gamma}^2 \tilde{E}_x}$$

wave equation for $\tilde{E}_x$ in the material medium.

Propagation constant

$$\bar{\gamma} \equiv \sqrt{j\omega\mu(\sigma + j\omega\varepsilon)} = \alpha + j\beta$$

Unit?

Attenuation constant

Phase constant

Solve simple PDE and re-write Phasors as Cosines

$$\frac{\partial^2 \tilde{E}_x}{\partial z^2} = \bar{\gamma}^2 \tilde{E}_x \qquad \boxed{\tilde{E}_x(z) = \bar{A} e^{-\bar{\gamma} z} + \bar{B} e^{+\bar{\gamma} z}} \qquad \bar{\gamma} = \alpha + j\beta$$

$$\tilde{E}_x(z) = \bar{A} e^{-\alpha z} e^{-j\beta z} + \bar{B} e^{\alpha z} e^{j\beta z}$$

$$= A e^{j\theta} e^{-\alpha z} e^{-j\beta z} + B e^{j\phi} e^{\alpha z} e^{j\beta z}$$

$$\vec{E}_x(z,t) = \mathrm{Re}[\tilde{E}_x(z) e^{j\omega t}]$$

$$\boxed{\vec{E}_x(z,t) = A e^{-\alpha z} \cos(\omega t - \beta z + \theta) + B e^{+\alpha z} \cos(\omega t + \beta z + \phi)}$$

$z > 0$ $\qquad$ $z < 0$

Also Solve for H_y

$$\vec{E}_x(z,t) = Ae^{-\alpha z}\cos(\omega t - \beta z + \theta) + Be^{+\alpha z}\cos(\omega t + \beta z + \phi)$$

z>0 z<0

$$\frac{\partial \tilde{E}_x}{\partial z} = -\mu(j\omega)\tilde{H}_y \qquad \overline{\eta} \equiv \sqrt{\frac{j\omega\mu}{\sigma + j\omega\varepsilon}} = |\overline{\eta}|e^{j\tau}$$

intrinsic impedance of the medium

$$\vec{H}_y(z,t) = \frac{A}{|\overline{\eta}|}e^{-\alpha z}\cos(\omega t - \beta z + \theta - \tau) - \frac{B}{|\overline{\eta}|}e^{+\alpha z}\cos(\omega t + \beta z + \phi - \tau)$$

z>0 z<0

Apply Boundary Conditions

$$\vec{E}_x(z,t) = Ae^{-\alpha z}\cos(\omega t - \beta z + \theta) + Be^{+\alpha z}\cos(\omega t + \beta z + \phi)$$

$$\hat{n}\times(\vec{\boldsymbol{E}}_1 - \vec{\boldsymbol{E}}_2) = 0$$

$\vec{\boldsymbol{E}}_t$ is continuous

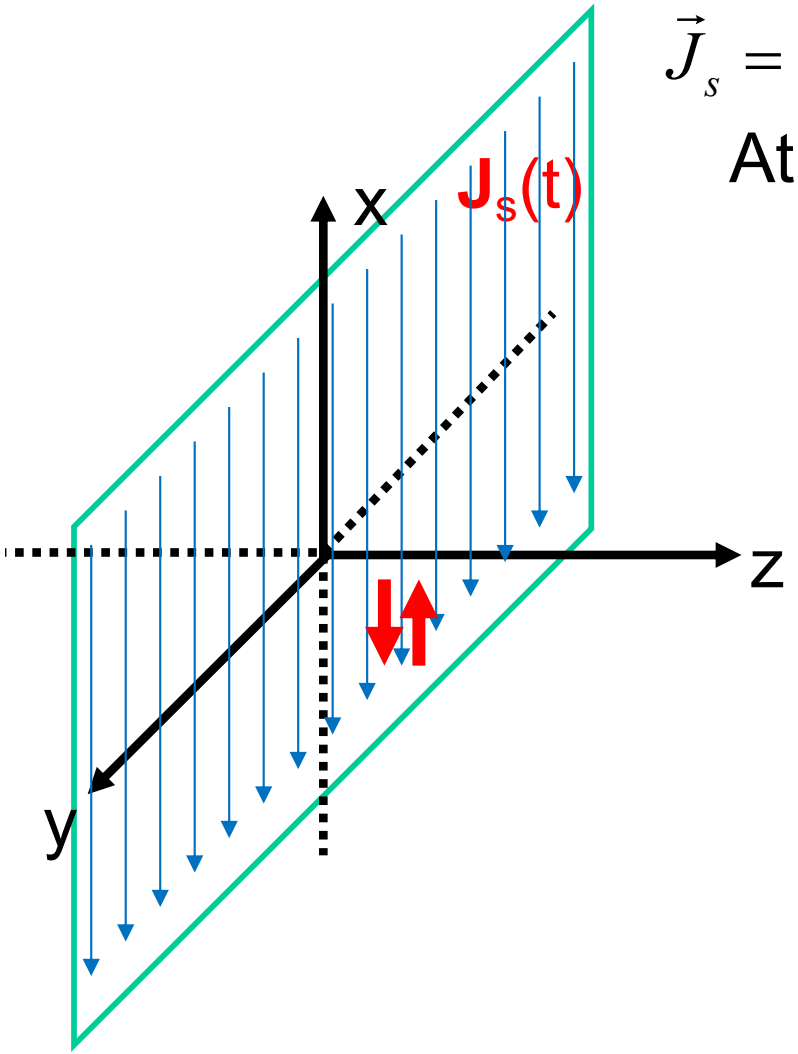
At z=0

$$A\cos(\omega t + \theta) = B\cos(\omega t + \phi)$$

$$\therefore A = B \text{ and } \theta = \phi$$

$$\vec{J}_s = -J_s(t)\hat{a}_x$$

At z=0



Apply Boundary Conditions

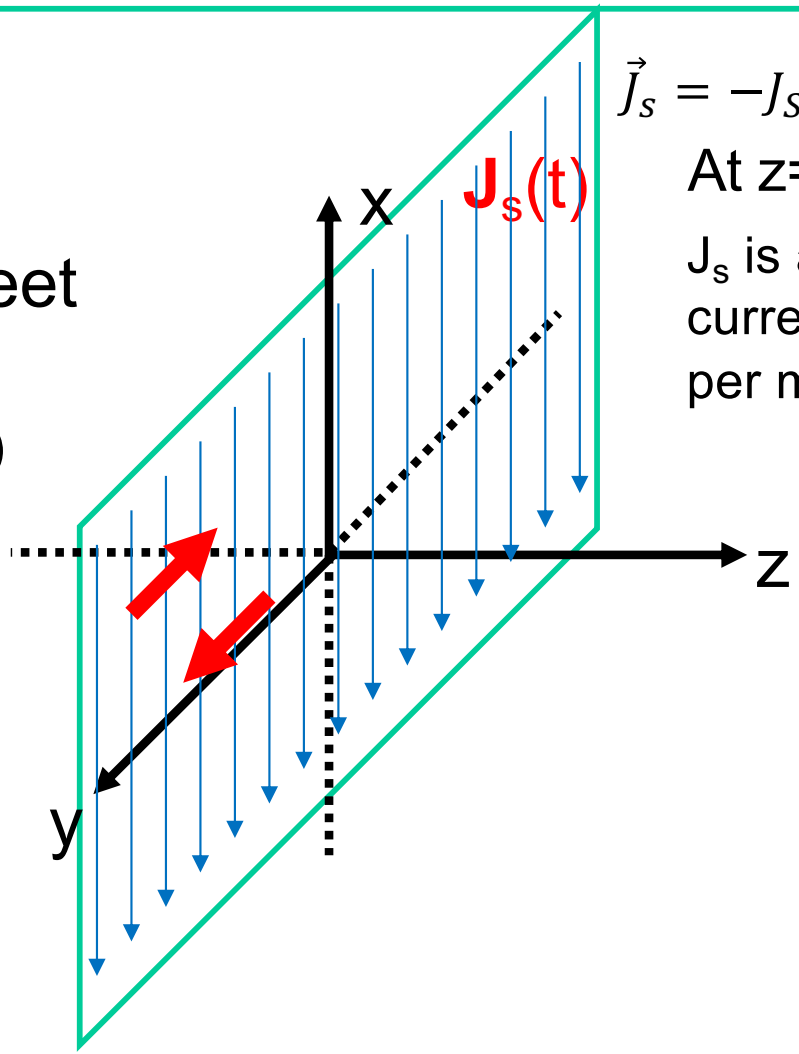
$$\vec{H}_y(z,t) = \frac{A}{|\overline{\eta}|} e^{-\alpha z} \cos(\omega t - \beta z + \theta - \tau) - \frac{A}{|\overline{\eta}|} e^{+\alpha z} \cos(\omega t + \beta z + \theta - \tau)$$

$$\hat{n} \times (\vec{H}_1 - \vec{H}_2) = \vec{J}_s$$

$\vec{H}_t$ is discontinuous at the sheet

$$\frac{2A}{|\overline{\eta}|} \cos(\omega t + \theta - \tau) = J_{S0} \cos(\omega t)$$

$$\therefore A = \frac{|\overline{\eta}| J_{S0}}{2} \text{ and } \theta = \tau$$



$\vec{J}_s = -J_{S0} \cos(\omega t) \hat{a}_x$

At z=0

J_s is a surface current in Amps per meter

Final Solution for E_x and H_y

$$\vec{E}(z,t) = \frac{|\overline{\eta}| J_{S0}}{2} e^{\mp \alpha z} \cos(\omega t \mp \beta z + \tau)\vec{a}_x$$

$$\vec{H}(z,t) = \frac{\pm J_{S0}}{2} e^{\mp \alpha z} \cos(\omega t \mp \beta z)\vec{a}_y \qquad z \gtrless 0$$

- Strength of fields drops exponentially according to the attenuation constant α
- Magnitudes |E| and |H| related through magnitude of the complex impedance, $|\overline{\eta}|$
- **E** and **H** are out of phase by the phase of the complex impedance, $\tau = \arg(\overline{\eta})$

Phasor Solution for E_x and H_y

$$J_x(z=0,t) = \mathrm{Re}[-\tilde{J}_{S0} e^{j\omega t}]$$

$$E_x(z,t) = \mathrm{Re}[\tilde{E}_x(z) e^{j\omega t}]$$

$$H_y(z,t) = \mathrm{Re}[\tilde{H}_y(z) e^{j\omega t}]$$

$$\tilde{E}_x(z) = \frac{\overline{\eta}\tilde{J}_{S0}}{2} e^{\mp\bar{\gamma}z} \hat{a}_x \qquad z \gtrless 0$$

$$\tilde{H}_y(z) = \frac{\pm\tilde{J}_{S0}}{2} e^{\mp\bar{\gamma}z} \hat{a}_y$$

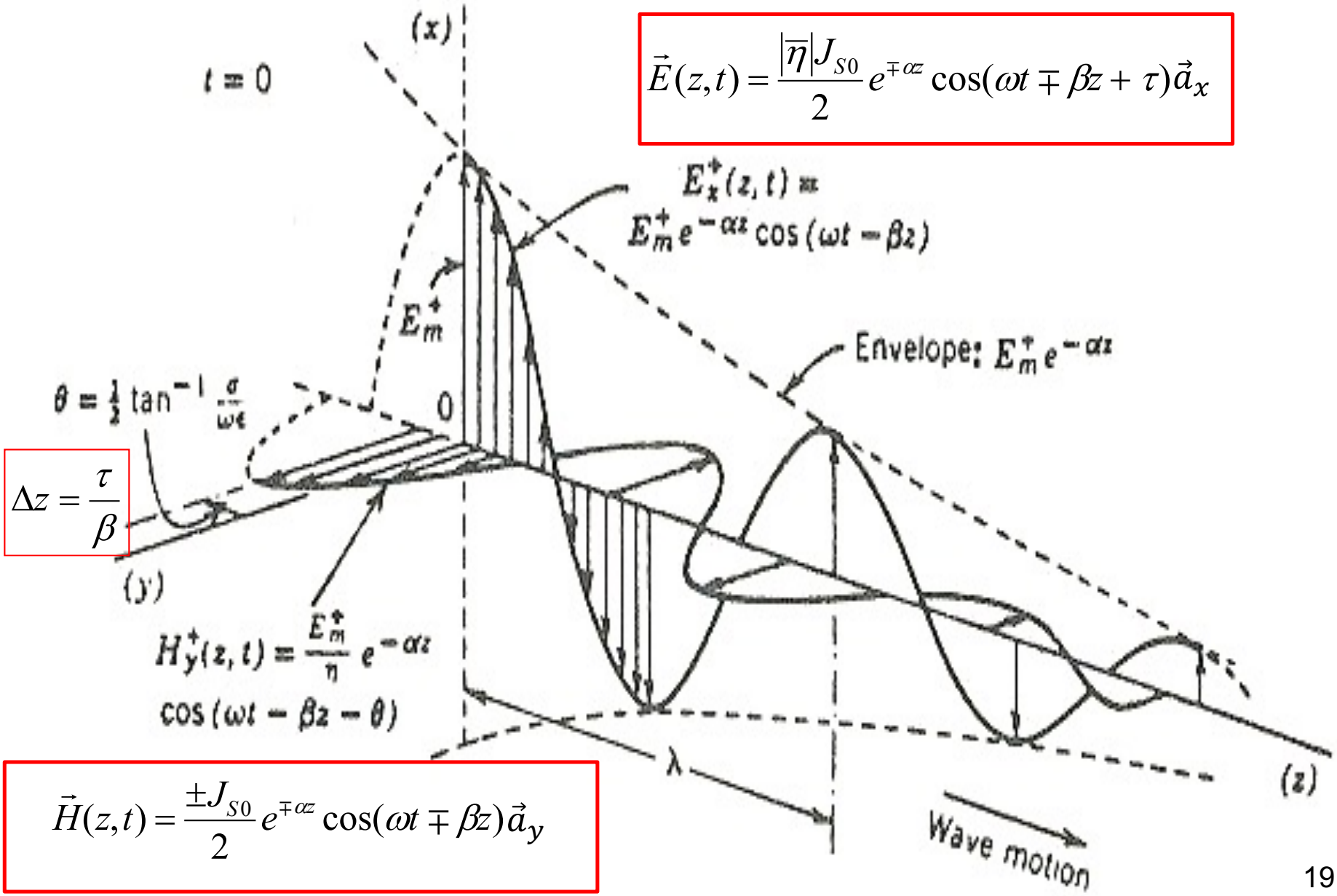
$\tilde{H}$ is in phase with the current source

$$"\tilde{E}_x = \overline{\eta}\tilde{H}_y"$$

holds for amplitude and for phase
but they point in different directions

In this drawing, the current sheet is not at z=0 Need to shift functions right by θ=τ for our case

$$\vec{E}(z,t) = \frac{|\overline{\eta}|J_{S0}}{2} e^{\mp\alpha z}\cos(\omega t \mp \beta z + \tau)\vec{a}_x$$



$$\vec{H}(z,t) = \frac{\pm J_{S0}}{2} e^{\mp\alpha z}\cos(\omega t \mp \beta z)\vec{a}_y$$

What is the same?

- Frequency of the wave = ω
- Perpendicular: **E** $\perp$ **H** $\perp$ **S**

$$\vec{S} = \vec{E} \times \vec{H}$$

What is different?

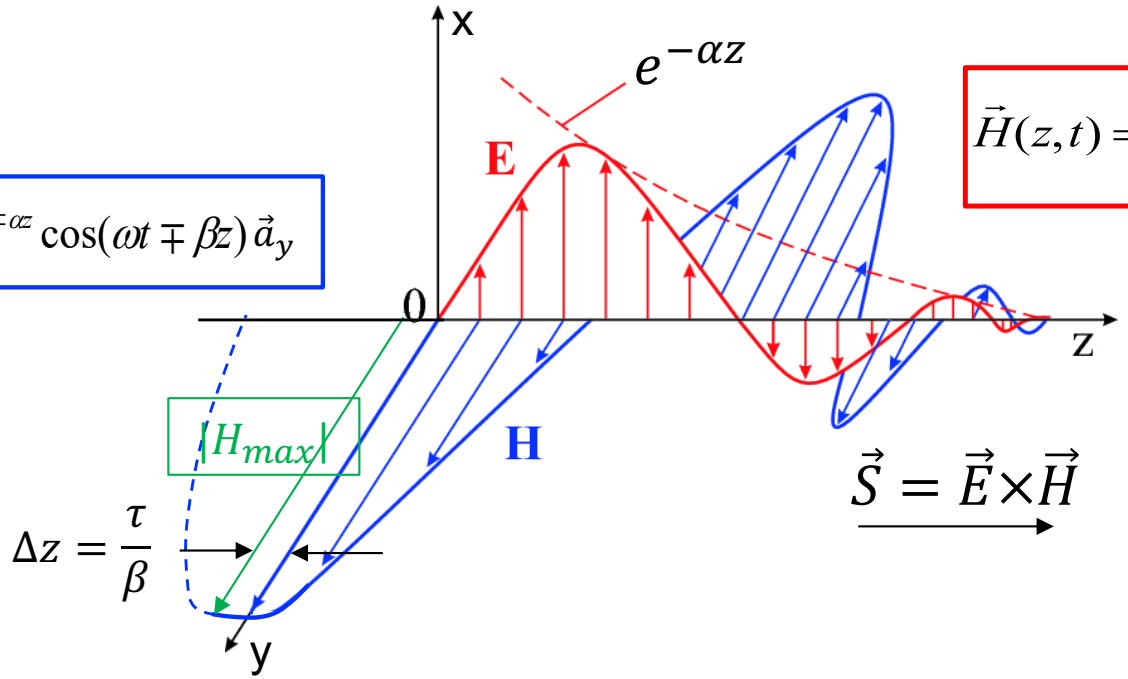
1. The magnitude of the wave is ATTENUATED

Gets weaker by $e^{-\alpha z}$ for a wave travelling in +z

α = "Attenuation Constant"
Units = 1/m or Np/m or dB/m

$$\vec{H}(z,t) = \frac{\pm J_{S0}}{2} e^{\mp \alpha z} \cos(\omega t \mp \beta z)\, \vec{a}_y$$

$$\vec{H}(z,t) = \frac{\pm J_{S0}}{2} e^{\mp \alpha z} \cos(\omega t \mp \beta z)\, \vec{a}_x$$



What is different?

2. Magnitude relationship between E and H

$$\left|\vec{E}\right| = \left|\overline{\eta}\right|\left|\vec{H}\right|$$

$$\left|\overline{\eta}\right| \neq \eta_0$$

$$\eta_0 = \sqrt{\frac{\mu_0}{\epsilon_0}}$$

$$\overline{\eta} = \sqrt{\frac{j\omega\mu}{\sigma + j\omega\epsilon}}$$

$\overline{\eta}$ = "Complex Impedance" of the material (because it is a complex number)

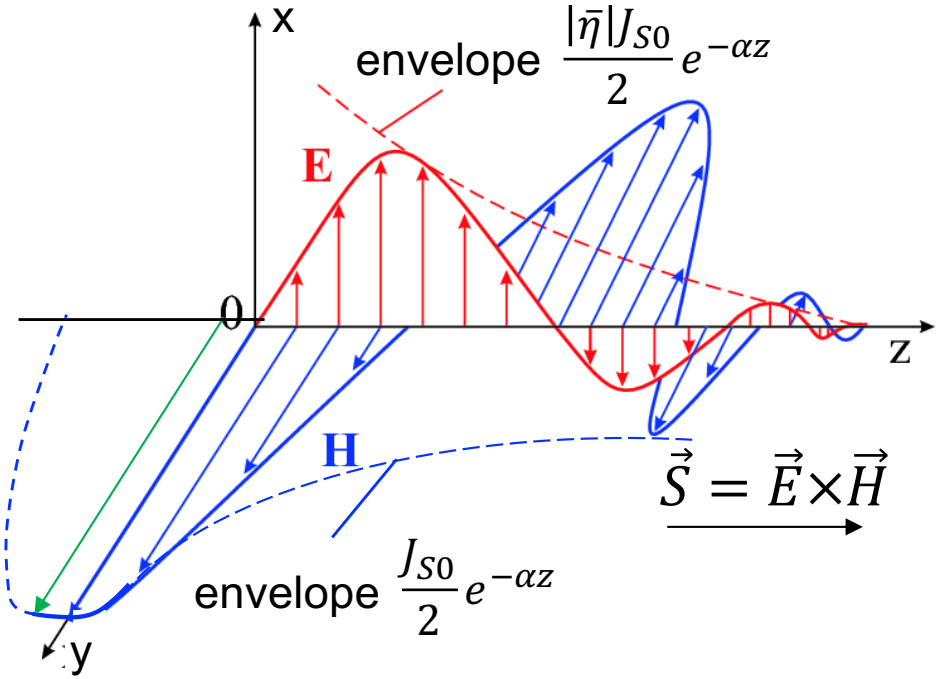
$\left|\overline{\eta}\right|$ = Magnitude of the complex impedance (Units = Ohms)

What is different?

3. **E** and **H** are OUT OF PHASE by τ $\quad \bar{\eta} = \sqrt{\dfrac{j\omega\mu}{\sigma + j\omega\epsilon}} = |\bar{\eta}|e^{j\tau}$

$$\tau = \angle\sqrt{\frac{j\omega\mu}{\sigma + j\omega\epsilon}}$$

"Phase Offset"
Units = radians



$$\vec{E}(z,t) = \frac{|\bar{\eta}|J_{S0}}{2}e^{\mp\alpha z}\cos(\omega t \mp \beta z + \tau)\vec{a}_y$$

$$\vec{H}(z,t) = \frac{\pm J_{S0}}{2}e^{\mp\alpha z}\cos(\omega t \mp \beta z)\vec{a}_y \qquad z \gtrless 0$$

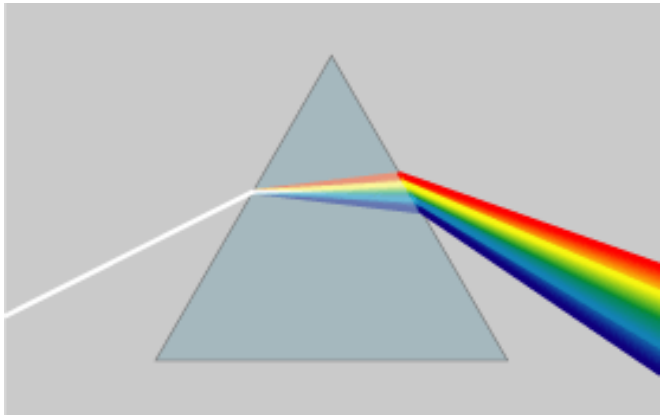
What is different?

4. Speed of propagation is no longer c in free space

$$v_p = \omega/\beta$$ Is still good

But β will be related to properties of the material and actually, β can depend on ω

If relation $\beta(\omega)$ is nonlinear, then the material has dispersion (different frequencies travel at different speeds)



colors separate in a prism due to dispersion

Challenge Question: Propagation in $\sigma = 0$ medium

Given the phasors: $\widetilde{E}_x(z) = \frac{\overline{\eta}\widetilde{J}_{S0}}{2} e^{\mp\bar{\gamma}z}\hat{a}_x \qquad \widetilde{H}_y(z) = \frac{\pm\widetilde{J}_{S0}}{2} e^{\mp\bar{\gamma}z}\hat{a}_y$

and definitions: $\overline{\eta} \equiv \sqrt{\frac{j\omega\mu}{\sigma + j\omega\varepsilon}} = |\overline{\eta}|e^{j\tau} \qquad \bar{\gamma} \equiv \sqrt{j\omega\mu(\sigma + j\omega\varepsilon)} = \alpha + j\beta$

If $\sigma = 0$ and $\epsilon = 2\epsilon_0$, consider the veracity of the statements:

I. The fields will attenuate as they propagate

II. **E** and **H** will be out of phase

III. The velocity of propagation will be c

(a) I only, (b) II only, (c) III only, (d) I, II, and III, (e) none are true

Lecture 21-22a Summary

- Differentiate Maxwell's Equations

$$\nabla \times \vec{E} = -\mu \frac{\partial \vec{H}}{\partial t} \qquad \nabla \times \vec{H} = \sigma \vec{E} + \varepsilon \frac{\partial \vec{E}}{\partial t}$$

to get complex wave equation: $\dfrac{\partial^2 \tilde{E}_x}{\partial z^2} = \bar{\gamma}^2 \tilde{E}_x$

which has decaying solutions:

$$\vec{E}(z,t) = \frac{|\bar{\eta}| J_{S0}}{2} e^{\mp \alpha z} \cos(\omega t \mp \beta z + \tau)\, \vec{a}_x \qquad \tilde{E}_x(z) = \frac{\bar{\eta} \tilde{J}_{S0}}{2} e^{\mp \bar{\gamma} z} \hat{a}_x$$

$$\vec{H}(z,t) = \frac{\pm J_{S0}}{2} e^{\mp \alpha z} \cos(\omega t \mp \beta z)\, \vec{a}_y \qquad \tilde{H}_y(z) = \frac{\pm \tilde{J}_{S0}}{2} e^{\mp \bar{\gamma} z} \hat{a}_y$$